Narrative approach and local projection

Qingyuan Fang [1]

[1] Johns Hopkins University

March 30, 2026

Overview

Big picture

We want an empirical object that behaves like a policy innovation:

$$\text{monetary policy shock}_t \approx \Delta i_t \perp\!\!\!\perp \mathcal{I}_t,$$

where $\mathcal{I}_t$ is the information set that drives systematic policy.

- Identification problem: endogeneity and anticipation.
- The narrative approach in Romer and Romer (2004):
  - Step 1: intended policy change around FOMC meetings.
  - Step 2: purge systematic responses to Fed forecasts.
- Implications: output & prices; why results differ from conventional measures.
- Bridge to local projections and what a modern LP replication would change.

Two core threats

Endogeneity (policy responds to the economy)
In booms, rates rise endogenously $\Rightarrow$ attenuate the contractionary estimate.

Anticipation (policy responds to Fed forecasts)
Even a "target" series embeds forward-looking responses, eg.

$$i_t = \phi\, \mathbb{E}_t[\pi_{t+1:t+h}, y_{t+1:t+h}] + \varepsilon_t.$$

If the Fed eases because it foresees a recession, output may fall *after* easing even when easing is stimulative.

Identification target: "deviations from the systematic rule"

What we want

A shock should be orthogonal to the information set driving the systematic component:

$$\varepsilon_t \perp\!\!\!\perp \mathcal{I}_t \quad \Rightarrow \quad \mathbb{E}[\varepsilon_t \mid \mathcal{I}_t] = 0.$$

What makes this hard in monetary policy

- $\mathcal{I}_t$ is huge: forecasts, judgment, informal signals, financial conditions, global news.
- Identification strategies differ mainly in how they approximate $\mathcal{I}_t$ and when they measure the surprise.

Preview: two complementary "information sets"

- Narrative approach: approximates *policymaker* information set.
- High-frequency approach: approximates *market* information set in a narrow window.

Why focus on FOMC meetings?

Key constraint

The staff forecast used as the proxy for $\mathcal{I}_t$ is produced for scheduled FOMC meetings.

T-6 days	Greenbook forecast; used heavily in deliberations.
T-5 to T	Bluebook alternatives; pre-meeting briefings; incoming data possibly triggers forecast update.
Meeting day	FOMC decisions shape intended stance.
Post-meeting	Desk implements operations; intermeeting moves may occur on new information.

Design choice

Concentrate on policy intentions over the window where the forecast is a good proxy.

Data Sources

Narrative sources

- Record of Policy Actions.
- Detailed meeting accounts.
- Bluebooks (Monetary Policy Alternatives): implementation + options.

Why narrative?

- Extends policy-intention measurement to periods with less explicit targets.
- Lets we distinguish: *policy decided at meeting* vs *reaction to new information*.

Quantitative internal source

Internal memos reporting the "expected federal funds rate" (Weekly Report of the Manager of Open Market Operations), used as an input/check for intended levels and changes.

Why focus on the funds rate?

- Explicit target for large parts of sample.
- Discussed even when reserves were emphasized.
- More consistent over time than quantity aggregates.

Step 1: Intended funds rate change around each meeting

Meeting-level object

For meeting m, infer two intended levels:

$ff_m^{\text{pre}} \equiv$ intended funds rate at forecast date, $ff_m^{\text{post}} \equiv$ intended funds rate implied by meeting decision.

Then

$$\Delta ff_m^I \equiv ff_m^{\text{post}} - ff_m^{\text{pre}}.$$

- Use narrative record to adjust for temporary deviations from intended levels.
- Combine narrative + expected-rate memos when timing is ambiguous:
  - Some near-post-meeting moves are *new information*; others were *pre-decided*.

Step 2

High-level specification (meeting-level)

$$\Delta ff_m^I = \alpha + \rho\, ff_m^{\text{pre}} + \underbrace{g\left(\widetilde{y}_{m,h}, \Delta\widetilde{y}_{m,h}, \widetilde{\pi}_{m,h}, \Delta\widetilde{\pi}_{m,h}, \widetilde{u}_{m,0}\right)}_{\text{Fed forecast controls}} + \varepsilon_m.$$

What goes into $g(\cdot)$ in the baseline

- Real output growth forecasts at horizons $h = -1, 0, 1, 2$ and changes since previous meeting.
- Inflation (deflator) forecasts at same horizons and changes since previous meeting.
- Current-quarter unemployment forecast.

Definition of the shock and how to interpret the residual

Shock definition

$$S_m \equiv \varepsilon_m \qquad \Rightarrow \qquad S_t \equiv \sum_{m \in \text{meetings in month } t} \varepsilon_m.$$

Interpret S_t as the change in intended policy not explained by forecast-based systematic responses.

Residuals are not "random mistakes"

Any regression-based shock absorbs everything not captured by the specification:

- operating procedures; beliefs; preferences; political pressure; other objectives; idiosyncratic factors;
- plus any missing information (if Greenbook is incomplete).

Sources of variation in S_t

- **Operating procedures**: reserve targeting episodes shift rates relative to forecast-implied rule.
- **Beliefs**: model of inflation-output tradeoff, NAIRU views, etc.
- **Preferences and tastes**: stronger inflation aversion $\Rightarrow$ higher rate for given forecast.
- **Politics**: pressure episodes can move the stance.
- **Other objectives**: e.g., exchange-rate concerns beyond forecast implications.
- **Idiosyncratic randomness**: personalities, persuasion, etc.

Effect on industrial production (Eq. 2)

$$\Delta y_t = a_0 + \sum_{k=1}^{11} a_k D_{k,t} + \sum_{i=1}^{24} b_i \Delta y_{t-i} + \sum_{j=1}^{36} c_j S_{t-j} + e_t.$$

Key design choices

- Use **non-seasonally adjusted** IP $\Rightarrow$ include month dummies $D_{k,t}$.
- Include many lags of Δy_t (normal dynamics) and S_t (policy lag structure).
- Exclude contemporaneous S_t: assume monetary policy affects output with at least a one-month lag.

How the impulse response is computed

A one-time 1pp realization of S maps into a path for Δy via the c_j and feedback through b_i; cumulate to get level effect on y.

Effect on prices (Eq. 3)

$$\Delta p_t = \alpha_0 + \sum_{k=1}^{11} \alpha_k D_{k,t} + \sum_{i=1}^{24} \beta_i \Delta p_{t-i} + \sum_{j=1}^{48} \gamma_j S_{t-j} + u_t .$$

Key takeaways

- Estimated price-level response is near zero for a long time (order of 2 years), then disinflation.
- Using broad measures like Δ funds rate can yield a price puzzle (prices rise after tightening).
- Adding commodity prices helps in some settings, but is not a complete fix if policy remains endogenous/anticipatory.

Why both corrections matter: "intended" and "forecast-purged"

Four instruments we should conceptually separate

1. Δi_t (*actual* funds rate change): mixes policy, noise, intermeeting info responses.
2. Δi_t^I (*intended* change around meetings): reduces short-run endogeneity, but still anticipatory.
3. residual from regressing Δi_t on forecasts: partial anticipation control, but keeps short-run endogeneity.
4. New measure: intended changes *and* forecast purge.

Empirical punchline to emphasize

Intermediate measures tend to look more like the broad measure than like the new measure
$\Rightarrow$ we need to address both endogeneity and anticipation to change estimated transmission meaningfully.

From Eq. (2) to local projections

Eq. (2) estimates one equation for Δy_t with long lag polynomials, then derives a multi-horizon response by iterating dynamics

Local projections (LP): horizon-by-horizon direct estimation

For each horizon $h = 0, 1, \ldots, H$ estimate a separate regression:

$$\Delta y_{t+h} = \alpha_h + \theta_h S_t + \Gamma_h' W_t + \nu_{t+h},$$

where W_t collects predetermined controls (lags of y, lags of S, and other covariates).

Narrative approach

More narrative datasets, clearer “instrument” framing

Narrative shocks are often best viewed as instruments (deviations from policy rules) used to trace causal propagation.

Scale narrative coding with text methods

- Use natural language from internal policy documents to proxy a broader information set than a few forecast series.
- Apply structured text/ML pipelines to extract and orthogonalize rich information embedded in documents.

Further Reading

Jordà, Ò. (2005). "Estimation and Inference of Impulse Responses by Local Projections." *American Economic Review*, 95(1), 161–182.

Ramey, V. A. (2016). "Macroeconomic Shocks and Their Propagation." In J. B. Taylor and H. Uhlig (eds.), *Handbook of Macroeconomics*, Vol. 2, pp. 71–162. Elsevier.

Aruoba, S. Boragan, and T. Drechsel (2024). "Identifying Monetary Policy Shocks: A Natural Language Approach." *NBER Working Paper* No. 32417.

Drechsel, T. (forthcoming). "Political Pressure on the Fed." *Review of Economic Studies*.

Fernandez-Fuertes, R. (2025). "Monetary Policy Shocks: A New Hope—Large Language Models and Central Bank Communication." *BAFFI Centre Research Paper* No. 257; SSRN Working Paper.

Martingale Difference Sequence (MDS)

A MDS e_t is defined with respect to a specific sequence of **information sets** $\mathcal{F}_t$.

Definition

The process $(e_t, \mathcal{F}_t)$ is a MDS if

- e_t is adapted to $\mathcal{F}_t$, meaning e_t is $\mathcal{F}_t$-measurable for all t.
- $\mathbb{E}\,|e_t| < \infty$
- $\mathbb{E}\,[e_t \mid \mathcal{F}_{t-1}] = 0$

Note: When no explicit definition is given it's standard to assume that $\mathcal{F}_t = \sigma(e_t, e_{t-1}, \ldots)$.

1. If e_t is i.i.d and mean zero, it's a MDS
2. If $(e_t, \mathcal{F}_t)$ is a MDS and $\mathbb{E}\left[e_t^2\right] < \infty$ then e_t is serially uncorrelated.

Ex. Show that if $(e_t, \mathcal{F}_t)$ is a MDS and X_t is $\mathcal{F}_t$-measurable then $u_t = X_{t-1}e_t$ is a MDS

Solution

u_t is $\mathcal{F}_t$-measurable; and

$$\mathbb{E}[u_t \mid \mathcal{F}_{t-1}] = \mathbb{E}[X_{t-1} e_t \mid \mathcal{F}_{t-1}]. = X_{t-1}\, \mathbb{E}[e_t \mid \mathcal{F}_{t-1}] = 0.$$